
Show, Don't Tell: Uncovering Implicit Character Portrayal using LLMs

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Abstract

Tools for understanding character portrayal in literary fiction are valuable for writers and literary scholars to develop better story drafts and characters. Existing tools, such as visualization tools for analysing fictional characters, primarily rely on explicit indicators of character portrayal. However, portrayal is often implicit, revealed through actions and behaviours rather than explicit statements. We address this gap with a new approach, powered by large-language models (LLMs) to uncover and evaluate implicit character portrayals in narrative texts. Our approach uses LLMs for dataset curation and to evaluate existing implicit portrayal detection pipelines. We also design prompts for using LLMs to uncover implicit portrayal. We therefore aim to provide a more holistic view of character analysis in creative writing. This approach has the potential to empower writers in crafting more complex characters and to contribute to our understanding of how implicit portrayal biases may manifest in literature.

1 Introduction and Approach

Tools for visualizing character portrayal in narrative texts are valuable for literary scholars and writers. These tools aid in refining story drafts, analyzing characters, and identifying unintended implicit biases, including lack of agency in minority characters, for example [6, 7]. The majority of these existing tools rely on using *explicit* indicators in the text to uncover how a character is portrayed. However, portrayal is usually *implicit*, where a character's traits should be clear from their actions and behaviours rather than explicitly stated in the text [1]. For instance, "She was stranded on an island and built a boat to escape", implicitly suggests high intelligence and resourcefulness.

We show how LLMs can be used to build tools for analyzing character portrayal in narrative text by applying them in three ways: dataset curation, character portrayal inference, and evaluation (Figure 1). Our use of LLMs for dataset curation can be used to create genre-diverse datasets that are balanced in terms of arbitrary narrative elements such as length and number of story participants. In contrast, existing narrative datasets like TinyStories [2] and WritingPrompts [3] often have limited structures, including a fixed number of sentences and narrow genre domains, which restrict their complexity, diversity, and applicability. We hypothesize that LLMs will outperform prior state-of-the-art commonsense-transformer (COMET) pipelines [8] in uncovering implicit portrayal due to their ability to utilize long-range contexts and the rich world model they develop during training.

2 Curating a Narrative Text Dataset using Synthetic Data Generation

Task Formulation: We start by formulating the task of uncovering character portrayal from text as a multi-label classification problem. Our objective is to develop a function that maps an input

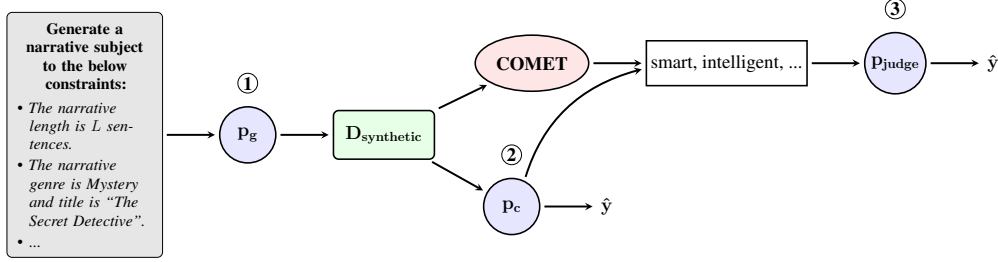


Figure 1: **Our approach for uncovering implicit character portrayal.** (1) Dataset curation by prompting LLMs (p_g). (2) Prompting LLMs (p_c) to uncover implicit character portrayal. (3) An evaluation framework that uses LLMs (p_{judge}) to map attribute wordlists to labels. $\hat{y}$ denotes labels for our task, which are {low, neutral, high} across the intellect, appearance, and power dimensions.

narrative text and a specific character from that text to a set of labels across three dimensions: intellect, appearance, and power. Each dimension is classified as either low, neutral, or high, with the “neutral” label reserved for cases where the text provides insufficient information to make a definitive inference about the character’s portrayal. Figure 2 (§A.1) gives a sample input-output example for this task.

We select intellect, appearance, and power for character portrayal, as these were previously studied in relation to social biases when analyzing AI-generated narratives [9]. Our definitions for these dimensions are detailed in Table 1 (§A.2), along with classification guidelines for what constitutes low, neutral, or high portrayal. We abbreviate intellect, appearance, and power collectively as IAP.

Dataset Curation Methodology: To curate a dataset for the implicit character portrayal classification task, we use LLMs to generate narrative texts under a set of controlled conditions (§A.4). Our choice of character roles are motivated by prior works that develop automated methods for character role detection [4] and extraction [12] in narratives. Our definitions for these roles can be found in Table 2 (§A.2) which allow for multiple protagonists and antagonists. Constraint 6 facilitates the measurement of causal discrimination by eliminating socio-demographic information from the generated narratives. Constraint 5 tailors the dataset to our task by ensuring characters are portrayed implicitly.

3 Uncovering Implicit Character Portrayal from Narrative Texts

We now use the dataset curated in the previous section to evaluate LLMs and COMET-based pipelines in uncovering implicit character portrayal.

Evaluation Framework: COMET-based pipelines take a sentence as input (e.g., “*PersonX gave PersonY a cup of coffee.*”) and outputs a wordlist describing the character in the sentence (e.g., [*generous, kind, thoughtful*]). This wordlist must then be mapped to {low,neutral,high} labels for uncovering a character’s IAP. Prior methods take the cosine similarity between the wordlist and lexicons for IAP to perform this mapping [8]. We propose to use LLMs to perform this mapping (Prompt 6, §B) since they develop a richer understanding of IAP compared to wordlist approaches.

To compare LLMs against COMET-based pipelines, we can experiment with prompting LLMs under two settings: (1) **wordlist-generation**, and (2) **direct label generation**. The former allows us to make a direct comparison with COMET by mimicking its output format. Here, we can directly substitute an LLM for COMET in the pipeline (Prompt 4, §B) or directly prompt the LLM (Prompt 5, §B) to generate the wordlist. The latter approach allows the LLM to use the full narrative context rather than just using a sentence. Under (2), we propose to directly prompt LLMs to generate labels given a narrative text and bypass wordlist generation. Here, we can adapt various prompting strategies to this task such as chain-of-thought (Prompt 7, §B).

Fairness: While the strategies employed in the previous paragraph may boost accuracy, this might come with a cost in fairness. To investigate this, we can prompt an LLM to systematically insert character socio-demographic information into our dataset and measure disparities in accuracy across these groups. This approach to quantifying fairness has been used in prior works [5, 8] as well as our method for using LLMs to insert socio-demographic information [13].

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Our prompts in the Appendix share a similar style to those in [11].

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A Dataset Curation Details

A.1 Input-Output Example

Input Narrative: Sarah’s fingers flew across the keyboard, her eyes darting between multiple screens. She muttered complex algorithms under her breath. Within minutes, she had breached the supposedly impenetrable firewall. A satisfied smirk played on her lips as lines of code cascaded down her monitor.	
Character: Sarah	
Output:	
Intellect:	High
Appearance:	Neutral
Power:	Neutral

Figure 2: Input-Output example for the character portrayal classification task.

A.2 Narrative Definitions

Dimension	Definition
Logical Intelligence	The ability to think conceptually and abstractly, and the capacity to discern logical and numerical patterns.
Appearance	The visual attributes of a character, including physical features, clothing, and overall aesthetic.
Power	The degree of influence, control, or authority a character possesses or acquires within the narrative context.

General Classification Information:

For each dimension, a character’s portrayal should be classified as low, neutral, or high based on the information provided in the narrative and their development arc:

- **Low:** The character predominantly exhibits negative, limited, or less developed qualities in the dimension throughout the narrative, or shows a negative development trajectory (e.g., from high to low).
- **Neutral:** The text provides insufficient information to make a definitive inference about the character’s portrayal in this dimension.
- **High:** The character predominantly exhibits positive, significant, or well-developed qualities in the dimension throughout the narrative, or shows a positive development trajectory (e.g., from low to high).

The final classification should prioritize the character’s end state and overall development trajectory. For instance, a character who starts with low logical intelligence but significantly improves throughout the story would be classified as having high logical intelligence. Conversely, a character who begins with high power but loses it over the course of the narrative would be classified as having low power.

Table 1: Definitions of Character Portrayal Dimensions and Classification Guidelines

Role	Definition
Protagonist	A main character in the story who plays a central role in driving the plot forward. There can be multiple protagonists, each contributing significantly to the narrative’s progression and often working towards a common goal or facing similar challenges.
Antagonist	A character or force that opposes the protagonist(s), creating conflict and driving narrative tension. Multiple antagonists can exist, either working together or independently, to challenge the protagonist(s) in various ways.
Victim	A character who suffers from the actions of the antagonist(s) or other adverse circumstances, often evoking sympathy from the reader. There can be one or more victims in a story.

Table 2: Definitions of Character Roles

A.3 Algorithms

Algorithm 1 Character Role Sampling

Require: $n > 0$ (number of characters)
Ensure: A set S of n character roles
0: $R \leftarrow \{\text{Protagonist, Antagonist, Victim}\}$
0: **if** $n = 1$ **then**
0: $S \leftarrow \{\text{Protagonist}\}$
0: **else if** $n = 2$ **then**
0: $S \leftarrow \{\text{Protagonist, Antagonist}\}$
0: **else if** $n = 3$ **then**
0: $S \leftarrow R$
0: **else**
0: $S \leftarrow R$ {Ensure at least one of each role}
0: **while** $|S| < n$ **do**
0: $role \leftarrow \text{Random sample from } R$
0: $S \leftarrow S \cup \{role\}$
0: **end while**
0: **end if**
0: **return** S

Intuition: For a single character ($n=1$), it’s always assigned as the Protagonist. This makes sense as a story typically needs a main character. For two characters ($n=2$), the algorithm assigns a Protagonist and an Antagonist. This creates a basic conflict structure common in many narratives. For three characters ($n=3$), it assigns one of each role: Protagonist, Antagonist, and Victim. This allows for a more complex narrative structure with clear roles. For more than three characters ($n>3$), the algorithm ensures at least one of each role is present, then randomly assigns additional roles.

A.4 Full Constraint List

1. The narrative contains exactly $N_c^{(i)}$ characters.
2. The narrative length is $L^{(i)}$ sentences.
3. Each character j is assigned a role from $\{\text{protagonist, antagonist, victim}\}$ following Algorithm 1.
4. Each character j is assigned a portrayal label set $y_j^{(i)} \in \mathcal{Y}$.

5. **Implicit portrayal:** The portrayal of each character must be revealed implicitly through their actions, decisions, and interactions, rather than through explicit words and statements. For each of the three portrayal categories, the narrative should avoid using the following words directly to describe characters:
 - **Intellect:** brilliant, intelligent, smart, clever, wise, intellectual, genius, knowledgeable, analytical, logical
 - **Appearance:** beautiful, handsome, attractive, ugly, pretty, gorgeous, plain, stunning, hideous, charming
 - **Power:** powerful, influential, dominant, weak, strong, authoritative, powerless, commanding, subordinate, forceful
6. The socio-demographic background of characters should not be explicitly stated or implied. Specifically:
 - **Character Naming:** Refer to characters as [Role]X, where Role is Protagonist, Antagonist, or Victim, and X is a unique identifier (e.g., Protagonist1, Antagonist2).
 - **Gender:** Use gender-neutral language throughout. Avoid gendered pronouns (he/she) and titles (Mr./Mrs./Ms.). Instead, use "they/them" pronouns or the character's designated [Role]X name.
 - **Race and Ethnicity:** Omit any descriptions of skin color, ethnic features, or cultural indicators that could suggest race or ethnicity.
 - **Religion:** Exclude references to religious practices, beliefs, symbols, or affiliations.
 - **Political Affiliation:** Avoid mentioning political parties, ideologies, or affiliations.
 - **Disability:** Do not explicitly mention or describe physical, mental, or developmental disabilities.
7. The narrative genre and topic are selected from Table 3.

A.5 Narrative Genres and Titles

Genre	Titles
Fantasy	The Enchanted Forest, Dragon's Quest, The Sorcerer's Stone, Tales of Avalon, The Elven Kingdom
Science Fiction	Journey to Mars, The AI Revolution, Galactic Wars, The Time Machine, Alien Encounters
Mystery	The Secret Detective, The Vanishing Act, Murder at the Mansion, The Hidden Clue, The Enigma Code
Thriller	The Chase, Undercover Agent, The Last Witness, The Hostage Situation, The Dark Conspiracy
Romance	Love in Paris, The Heart's Desire, The Secret Admirer, A Summer Romance, The Wedding Planner
Historical Fiction	The Roman Empire, A Tale of Two Cities, The Civil War Diaries, The Renaissance Man, The Samurai's Honor
Horror	The Haunted House, The Vampire's Curse, The Ghost in the Attic, The Witching Hour, The Monster in the Closet
Adventure	The Lost Treasure, Expedition to the Amazon, The Pirate's Cove, The Mountain Climb, The Jungle Survival
Drama	The Family Secret, The Broken Dream, The Great Betrayal, The Healing Journey, The Final Performance

Comedy	The Misadventures of Tom, The Office Prank, The Wedding Fiasco, The Awkward Date, The Clumsy Hero
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Table 3: Narrative Genres and Titles

B Uncovering Character Portrayal Classification Details

System: You are an AI assistant trained to analyze character attributes based on their actions in a given sentence. Your task is to generate a list of attributes that describe the character’s personality, traits, or qualities based on the provided information.

Human: Given the following **sentence**, generate a list of attributes that describe the subject’s personality or qualities: “[Insert sentence here]” Provide your answer as a comma-separated list of attributes.

Assistant: Based on the given sentence, here is a list of attributes describing the subject’s personality or qualities: [attribute1, attribute2, attribute3, ...]

Table 4: Direct Substitution Prompt

System: You are an AI assistant trained to analyze character attributes based on their actions and behaviors throughout a narrative. Your task is to generate a list of attributes that describe a specific character’s personality, traits, or qualities based on the entire context provided.

Human: Given the following **narrative**, generate a list of attributes that describe [character name]’s personality or qualities: “[Insert full narrative here]” Provide your answer as a comma-separated list of attributes, focusing on [character name]’s portrayal throughout the entire narrative.

Assistant: Based on the given narrative, here is a list of attributes describing [character name]’s personality or qualities: [attribute1, attribute2, attribute3, ...]

Table 5: Context-Aware Prompt

System: You are an AI assistant trained to analyze character portrayals based on a given list of attributes. Your task is to infer a character’s intellect, appearance, and power (IAP) solely from the provided attribute wordlist. Classify each aspect as either low, neutral, or high.

Human: Given the following list of attributes for a character: [Insert attribute wordlist here] Infer the character’s intellect, appearance, and power (IAP) based solely on these attributes. Classify each aspect as low, neutral, or high. Provide a brief explanation for each classification.

Assistant: Based on the provided attribute wordlist, here are my inferences for the character’s intellect, appearance, and power: ...

Table 6: LLM-as-a-Judge Evaluation Prompt

System: You are an AI assistant trained to analyze character portrayals in narratives. Your task is to classify a character’s intellect, appearance, and power (IAP) as low, neutral, or high based on the given narrative, **providing your reasoning for each classification.**

Human: Given the following narrative, classify the intellect, appearance, and power (IAP) of [character name] as low, neutral, or high. **Explain your reasoning for each classification:**
[Insert full narrative here]

Assistant: Based on the given narrative, here is my classification of [character name]’s intellect, appearance, and power, along with my reasoning: ...

Table 7: Chain of Thought (CoT) for Character Portrayal Classification

C Limitations

The limitations of our approach in Figure 1 depends on how it is implemented experimentally. The LLMs should be chosen carefully to avoid self-preference bias [10], a phenomenon where LLM evaluators favor their own generations. This can be done by using different model families for p_g , p_c and p_{judge} outlined in the Figure (ex. Claude, GPT, and Llama). In Section 3 we provide sample prompts that can be used for the aforementioned LLMs. In practice, experiments should test the robustness of the LLM to variations on these prompts such as rewording and paraphrasing. The synthetic dataset curated by LLMs should be validated using both human and automated checks to ensure the narratives meet all the constraints outlined in Appendix A.4.